Prabhavit Gupta

AI Engineer | Generative AI, LLM Pipelines, Model Serving & Cloud Architecture

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Executive Summary

- **Generative AI & Machine Learning Solutions:** AI Engineer with 3 years commercial software experience developing, evaluating, and deploying enterprise Generative AI systems, conversational assistants, and automated data pipelines across Italy and Europe.
- **Production Model Serving & Performance Optimization:** Extensive expertise building high-throughput, low-latency microservices with FastAPI, Docker, and asynchronous task queues, optimizing inference runtime and semantic caching across cloud environments (GCP/AWS).
- **Local Presence & Italian Market Alignment:** Based in Venice/Padua (Veneto), holding an EU Permanent Residence Permit (Permesso di Soggiorno a Tempo Indeterminato) with unrestricted Italian working rights, working Italian language capability, and immediate availability for hybrid collaboration in Milan or Padua.

Technical Skills & Core Competencies

- **Generative AI & LLM Engineering:** LangChain, LangGraph, Model Context Protocol (MCP), Hugging Face Transformers, vLLM, OpenAI API, Anthropic Claude, Google Gemini, Prompt Optimization, RAG Architectures.
- **Machine Learning & Data Science:** Scikit-Learn, PyTorch, Pandas, NumPy, Data Pipeline Engineering, Feature Extraction, Embedding Models, Semantic Search, Evaluation Frameworks (DeepEval, RAGAS).
- **Backend & Model Serving:** Python 3.11+, FastAPI, RESTful API Design, Asynchronous Programming (asyncio), Celery, Redis, Pydantic Schema Validation.
- **Databases & Vector Stores:** PostgreSQL, pgvector, Qdrant, Pinecone, ChromaDB, SQL Query Optimization, Redis Cache.
- **Cloud, DevOps & Tooling:** Google Cloud Platform (Cloud Run, Vertex AI, Pub/Sub), AWS (ECS, Lambda), Docker, Git, GitHub Actions CI/CD, Linux / Bash, Prometheus / Grafana Monitoring.
- **Methodologies & Governance:** Agile / Scrum, Enterprise IT Architecture, Data Governance (EU GDPR Compliance), Automated Testing (pytest).

Professional Experience

AI Engineer & Consultant • *Self-Employed / Independent AI Consultancy*
Oct 2024 – Present • Venice / Hybrid Italy

- Designed and shipped end-to-end Generative AI applications and automated document intelligence pipelines for corporate clients, integrating open-source LLMs and proprietary foundation models into core workflows.
- Architected high-performance RAG pipelines combining dense vector embeddings with sparse keyword search (BM25) over unstructured enterprise knowledge repositories, achieving 94% contextual retrieval accuracy.
- Deployed scalable FastAPI microservices containerized with Docker on cloud platforms, implementing asynchronous worker pools (Celery/Redis) and sub-200ms latency endpoints.
- Built automated regression and safety evaluation harnesses utilizing golden test sets, monitoring model hallucination rates, token economics, and GDPR data masking prior to production rollouts.
- Provided technical consulting and architecture reviews to cross-functional teams, translating business automation requirements into robust technical specifications and measurable ROI.

AI Software Engineer • *Freelance & Client Implementations*
Jan 2023 – Sep 2024 • Veneto / Italy

- Developed and maintained data ingestion, cleansing, and validation pipelines processing multi-source structured and unstructured data for enterprise analytics and ML modeling.
- Implemented semantic search engines and conversational AI agents with tool-calling capabilities, connecting foundation models to internal SQL databases and external REST APIs.
- Optimized LLM prompt templates and token consumption strategies, reducing API inference expenditures by 40% while preserving output quality and schema compliance.
- Containerized production services using Docker and orchestrated automated deployment pipelines with GitHub Actions, ensuring consistent staging and production parity.
- Maintained strict data privacy protocols in full adherence with EU GDPR regulations, enforcing automated PII redaction on all inbound model prompts.

Featured AI Systems & Engineering Projects

Enterprise Retrieval-Augmented Generation Platform • *Lead Engineer*
Production Implementation • 2025

- Architected an enterprise-grade knowledge discovery platform leveraging hybrid search (pgvector + BM25), reciprocal

rank fusion (RRF), and Cohere rerankers across 100k+ technical documents.
- Built a streaming FastAPI serving layer with Redis semantic caching, reducing median query response time from 1.8s to 280ms under concurrent traffic loads.

Autonomous Agentic Tool Integration Framework • *Creator*

Open-Source / Production Reference • 2024 – 2025

- Implemented an agentic orchestration layer based on the Model Context Protocol (MCP), enabling autonomous agents to safely inspect schemas, query databases, and execute validated API calls.
- Included comprehensive telemetry and structured tracing via OpenTelemetry, enabling end-to-end visibility into agent reasoning trajectories and tool execution results.

Education, Credentials & Languages

- **Bachelor of Arts in Philosophy & Economics** • Ca' Foscari University of Venice (Coursework in Formal Logic, Decision Theory & Quantitative Systems)
- **Languages:** English (C1 – Full Professional Proficiency), Italian (Working Proficiency / Living in Italy), Hindi (Native)
- **Work Authorization:** EU Permanent Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) • Full, unrestricted right to work in Italy (CCNL Commercio or Freelance/P.IVA ready) • Located in Venice/Padua area.